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Predictors of early recurrent seizure after first seizure presentation to an emergency service: A retrospective cohort study

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ABSTRACT

Purpose: To identify the predictors of seizure recurrence after a first seizure (FS) presentation to an emergency service.
Methods: The clinical characteristics of FS patients presenting to the emergency department (ED) of our university hospital from January 2001 to December 2014 were retrospectively reviewed. Recurrence of seizures following the FS was classified as: early recurrence (0–1 month), intermediate recurrence (> 1–3 months), and late recurrence (> 3–12 months). The significant predictors of seizure recurrence in each period were identified by Cox proportional hazard ratios ($p < 0.05$).
Results: 648 FS patients of the 1248 overall seizures patients were initially enrolled. 414 FS patients were eligible for statistical analysis. Following the FS, 134 patients (32.4%) had recurrent seizures in which half of the first recurrences occurred within 3 months. The significant predictors of overall recurrence were remote symptomatic seizure (RSS) (adjusted hazard ratio [adj. HR] = 2.21 (1.38, 3.55), $p = 0.003$) and nocturnal onset seizure (NOS) (adj. HR = 1.53 (1.03, 2.26), $p = 0.039$). Those of early recurrence were NOS (adj. HR = 2.78 (1.44, 5.37), $p = 0.002$) and anti-epileptic drug (AED) prescription (adj. HR = 2.19 (1.00, 4.80), $p = 0.038$). Those of intermediate recurrence were RSS (adj. HR = 3.96 (1.63, 9.60), $p = 0.006$) and AED prescription (adj. HR = 4.90 (1.42, 16.95), $p = 0.003$). No predictor of late recurrence was identified.
Conclusion: The seizure recurrence rate was high in the first 3 months following the FS. The significant predictors were RSS, NOS and AED prescription.

1. Introduction

Approximately 5–10 % of the general population will experience a seizure in their lifetimes [1–3]. However, only 2–3 % of them will have one or more recurrences leading to a diagnosis of epilepsy. The challenge for proper management of a first seizure (FS) is to differentiate between epileptic and non-epileptic FS, upon which the decision on prescribing an anti-epileptic drug (AED) will be based. There are several studies with various methods and settings such as community-based, hospital-based, or specific conditions-based studies, which have reported different rates and predictors of recurrent seizures after a FS. Generally, 21–45 % of patients who presented with a FS would have a recurrent seizure within 2 years, and the first recurrent seizures mostly occurred within the first 3–6 months after the FS in most of the community-based studies [4–8]. The significant predictors of seizure recurrence after a FS included the presence of associated neurological abnormalities (*i.e.* congenital or acquired cerebral insults), remote symptomatic seizure (RSS), nocturnal onset seizure (NOS), and clinically-relevant abnormalities in cerebral magnetic resonance imaging (MRI) and/or electroencephalogram (EEG). Patients who experienced FS associated with these factors were estimated to have a two-fold higher risk of recurrence (70 %) [9,10].

Due to the limited number of studies elucidating seizure recurrence after a FS under the setting of emergency care, we aimed to investigate the significant predictors of seizure recurrence from the presenting characteristics of FS in an emergency service.

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2. Material and methods

2.1. Study population and setting

We retrospectively reviewed the demographic and clinical data of patients aged ≥15 years who presented to the ED of our institutional hospital with FS from January 2001 to December 2014, and were admitted for at least 48 h for clinical observation after initial treatment. We excluded cases with a known history of epilepsy and all epilepsy mimics (*e.g.* syncope, transient ischemic attack, psychogenic non-epileptic seizure, and extrapyramidal movement disorders) following consensus of our neurologist team. The hospital is a 900-bed, medical teaching university hospital, and the single tertiary-level medical center in southern Thailand.

2.2. Definitions of terms

We defined the terms used in this study as follows: a) *First seizure* (FS) refers to a seizure disorder that was reported by the patient or relative as the first-ever seizure in the patient's lifetime; b) *Daytime onset seizure* (DOS) is a seizure that took place during the daytime, while *nocturnal onset seizure* (NOS) is one that occurred during sleep at night; c) *Focal* and *generalized seizure* are types of seizure onset observed by a witness or self-reported by the patient at the presentation to the ED; d) *Status epilepticus* is an episode of multiple seizures or a single seizure without regaining of consciousness during the inter-ictal or post-ictal periods; and e) *Recurrent seizure* (RS) is a second seizure that occurred more than 24 h after the FS had been successfully treated. In addition, the etiologies of seizure were classified as: 1) *acute symptomatic seizure* (ASS) is a seizure that occurred within a week of an acute central nervous system (CNS) or systemic illness that secondarily affected the brain physiology, 2) *remote symptomatic seizure* (RSS) is a seizure in the presence of a history of known previous CNS insult, presumed to result in a static encephalopathy associated with an increased risk for epilepsy, and 3) *cryptogenic seizure* (CGS) is a seizure without an identifiable cause.

2.3. Data collection

Demographic data, clinical characteristics of the FS, associated comorbidities, and the results of initial investigations at the ED were collected for the study. The details of clinical characteristics of the FS included age, gender, concurrent illnesses and medications, family history of epilepsy, onset time of the FS (daytime *vs*. nocturnal), seizure classification (focal, generalized seizure disorder, unknown onset, or status epilepticus), seizure etiology (acute or remote symptomatic, or cryptogenic), duration of the FS, results of initial blood chemistry assessments, brain imaging studies (computed tomographic [CT] scan or magnetic resonance image [MRI] of the brain as available), and AED prescription based on the clinical assessment of the risk of seizure recurrence by a neurologist at the time of discharge from the ED. Patients aged ≥ 65 years were classified as "elderly". After the FS patients were discharged from the ED, they were re-evaluated and followed by a neurologist at the outpatient department until discontinuation from scheduled follow-ups or there was no indication for continuing treatment. We performed a retrospective cohort study of the enrolled cases. The timing of the first recurrent seizures was recorded.

The overall incidence of the first recurrent seizures and significant independent predictors of recurrence were initially assessed. We then identified the significant predictors of early occurrence of the recurrent seizures after the FS. The timing of seizure recurrence was classified as: early recurrence (0–1 month), intermediate recurrence (> 1–3 months), and late recurrence (> 3–12 months).

2.4. Statistical analysis

Descriptive statistics were used to demonstrate significant differences in the baseline demographics and presentation characteristics between the seizure recurrence and recurrence-free groups. Significant predictors for recurrent seizures were identified using Cox proportional hazards models over the total follow-up time, as well as in each predetermined interval of the occurrence of the first recurrent seizures noted above.

2.5. Ethical considerations

The study protocol was reviewed and approved by the Ethics Committee of the Faculty of Medicine, Prince of Songkla University (Registration no. 58-333-14-4). We strictly followed the 1964 Declaration of Helsinki, its later amendments, and all the current good practice guidelines in conducting the research. The identifiable personal information of all patients was completely anonymized. We have read the ethical and related guidelines for submission of a manuscript of this journal, and affirm that this manuscript is consistent with those guidelines. No grant or any other support was received for the study and writing of this manuscript. All authors declare they have no conflicts of interest involved with this study.

3. Results

3.1. General demographic and characteristics of first seizure

A total of 1248 patients who presented to the ED with seizure disorders during the study period were initially identified. Among them were 648 patients diagnosed with FS, and 414 (33.2%) cases were subsequently eligible for statistical analysis. The studied cases were 267 (64.5 %) males and 147 (35.5 %) females. Excluded from the analysis were 572 (45.8 %) cases with a known history of seizure disorders, 28 (2.2 %) with epilepsy mimics (16 with syncope, 4 with transient ischemic attack, 3 with psychogenic non-epileptic seizure, and 5 with extrapyramidal symptoms), 171 (13.7 %) patients who did not complete their follow-up evaluations after ED discharge, and 63 (5.0 %) cases with incomplete data records (Fig. 1). The median (IQR) age of the patients was 50.5 (32.0, 66.0) years. 147 (35.5 %) cases were elderly (aged ≥ 65 years). The major presenting seizure types were generalized seizure in 217 (53.8 %) cases, followed by 142 (35.2 %) cases of focal seizure, 44 (10.9 %) cases of status epilepticus, and 7 (1.7 %) cases of unknown seizure type. Two hundred and sixty cases (62.8 %) had daytime onset seizures (DOS) while 131 (31.6 %) cases had nocturnal onset seizures (NOS). Nearly half of the identifiable seizure etiologies were ASS (183 cases, 44.2 %) while 119 (28.7 %) and 112 (27.1 %) cases were RSS and CGS, respectively. Abnormal CT scans or MRIs of the brain or both presumably relevant to the epileptogenic lesions were found in 228 (61.3 %) cases (Table 1). EEGs were done after discharge from the ED in 35 (8.5 %) cases, of which 18 cases demonstrated epileptiform discharges.

3.2. Recurrent seizures: incidence and predictors

Overall, one or more recurrent seizures occurred in 134 of the 414 study cases (32.4 %) during the median (IQR) follow-up time of 33.4 (6.3, 66.1) months. The timing and number of cases with first recurrent seizure were: 0–1 month 38 (28.4 %); > 1–3 months 30 (22.4 %); > 3–12 months 33 (24.6 %); > 12–24 months 25 (18.7 %); and > 24 months 8 (6.0%). Only RSS, NOS, AED prescription, and history of stroke were significant factors associated with seizure recurrence in the univariate analysis (Table 1). However, only RSS (adjusted hazard ratio [adj. HR] = 2.21, 95 % CI 1.38–3.55) and NOS (adj. HR = 1.53, 95 % CI 1.03–2.26) were significant predictors of overall seizure recurrence after adjusting for all demographic and clinical

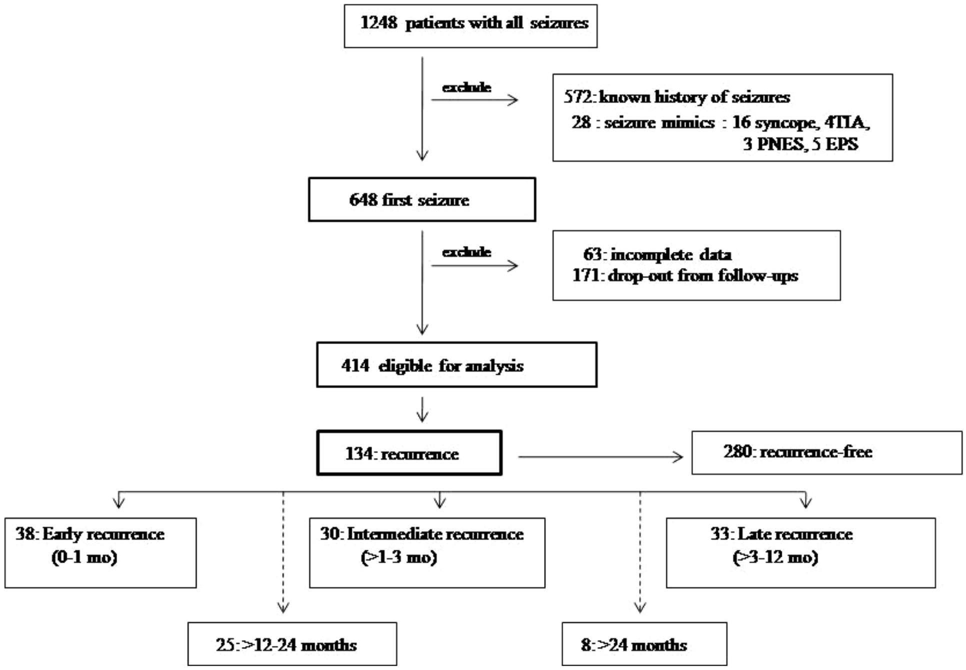

Fig. 1. Diagram illustrating the number of cases enrolled in this study.
Abbreviations: TIA: transient ischemic attack, PNES: psychogenic non-epileptic seizure, EPS: extrapyramidal symptoms.

characteristics of FS. When first recurrent seizures in each period were considered, NOS and AED prescription on discharge were significant predictors of early recurrence (0–1 month). RSS and AED prescription were significant predictors in the intermediate recurrence group (> 1-3 months). We found older age was a protective factor during this period of recurrence. No significant predictor could be determined in the late recurrence group (> 3–12 months) (Table 2).

4. Discussion

The study found that 32.4 % (134/414) of the eligible-for-analysis patients who presented to the ED with FS had one or more recurrent seizures. A review article of first unprovoked seizures reported a wide range of incidences of recurrent seizures (27–52%) after the initial seizure [11]. In general, a number of studies concluded the overall recurrence rate is approximately 40 % after FS (both provoked and unprovoked seizures) [3,5,8,12,13]. Also, a small prospective study found that 44.6 % of the patients who presented to an ED with FS had recurrent seizures at a mean time of 25.22 (± 13.69) months following the FS [14].

Based on the available evidence as discussed above, it can be estimated that around 40 % of patients with FS should be diagnosed with epilepsy at their first presentations [15,16]. The explanation for this clinical remark is that the apparent FS is not necessarily the first-ever seizure in the patient's lifetime. Previous subtle or non-convulsive epileptic spells could have been unrecognized by the patients, or underestimated as a non-specific transient neurological event by their physicians, causing appropriate investigations for seizure to be overlooked.

The rate of first recurrent seizure in the current study was found to be highest during the first month (28.4 %) followed by the > 1–3-month interval (22.4 %). The rate then declined to 11–18 % during the rest of the first year, and eventually lowered to 6 % beyond the second year. Unlike our study, most of the available studies in the literature included only cases with unprovoked seizure. A recent evidence-based guideline for the management of first unprovoked seizure stated that the prevalence of recurrence was highest within the first two years after the FS (21–49 %) [8]. Two large community-based epilepsy cohort studies reported on this topic. The FIR.S.T study reported accumulative recurrence rates of 18 %, 28 %, 41 %, and 51 % at 3, 6, 12, and 24 months, respectively, while the MESS study reported accumulative recurrence rates of 26 %, 39 %, 51 %, and 52 % at 6 months, 2, 5, and 8 years, respectively. Notably, 50.8 % of all recurrent seizures in our study occurred earlier within the first three months after the FS in our ED service setting. We believe that the higher severity and larger proportions of both recent and remote symptomatic seizure etiologies, which are possibly epileptogenic in our ED patients, contribute to the earlier and higher rate of seizure recurrence in our study. Therefore, it is crucial to closely monitor for seizure recurrence after FS patients are discharged from an ED.

RSS and NOS, but not older age as previously mentioned, were significant predictors of overall recurrent seizures in our study. We found that NOS and AED prescriptions were significant predictors of early recurrence (0–1 month), while RSS and AED prescriptions were significant in intermediate recurrence (> 1–3 months), but no significant predictors could be identified in late recurrence (> 3–12 months). In contrast to the previous studies, our study found that the elderly patients (aged ≥ 65 years) had significantly less risk of seizure recurrence during the intermediate recurrence phase (Table 2). It has been well established that RSS is a risk for seizure recurrence because the previous cerebral insults are epileptogenic in nature. Furthermore, most of the NOS cases included in the previous studies were focal onset seizures, particularly frontal and temporal lobe epilepsy, which had a high tendency to recur during non-rapid eye movement sleep [17–20]. Therefore, NOS could be considered as another risk factor of seizure recurrence due to its focal-onset seizure type. One prospective cohort study reported that the elderly had a higher percentage of seizure recurrence compared to younger group (53 % [95 % CI 46–62] *vs.* 48 % [95 % CI 44–51]). However, the following multivariate analysis showed that the independent predictors for seizure recurrence were RSS, NOS, epileptiform discharges on EEG and partial onset seizure, while advanced age was not found to be a significant predictor [21]. Other studies also reported that patients who presented with FS and had associated neurological deficits, abnormalities in neuroimaging studies (CT scan or MRI of brain, or both) or epileptiform discharges in EEG (spikes, slow waves or both) had a 70 % risk of seizure recurrence

Table 1
Demographic data and clinical characteristics of first seizure patients comparing between those with seizure recurrence and no recurrence after the first seizure.

Variable	Total N = 414	No seizure recurrence N = 280	Seizure recurrence N = 134	*P* value
Demographic data and characteristics of first seizure				> 0.999
Gender				
Female	147 (35.5)	99 (35.4)	48 (35.8)	
Male	267 (64.5)	181 (64.6)	86 (64.2)	
Age				0.524
median (IQR)	50.5 (32.0, 66.0)	51.5 (33.0, 67.0)	49.0 (31.2, 64.8)	
Age group				0.986
< 65 y	267 (64.5)	180 (64.3)	87 (64.9)	
≥ 65 y	147 (35.5)	100 (35.7)	47 (35.1)	
Family history of epilepsy				> 0.999
No	138 (98.6)	81 (98.8)	57 (98.3)	
Yes	2 (1.4)	1 (1.2)	1 (1.7)	
Seizure type				0.273
Focal	142 (34.6)	99 (35.7)	43 (32.3)	
Generalized	217 (52.9)	139 (50.2)	78 (58.6)	
Status epilepticus	44 (10.7)	33 (11.9)	11 (8.3)	
Unknown	7 (1.7)	6 (2.2)	1 (0.8)	
Duration of seizure				0.390
1–5 min	207 (56.7)	144 (58.5)	63 (52.9)	
6–10 min	94 (25.8)	58 (23.6)	36 (30.3)	
> 10 min	64 (17.5)	44 (17.9)	20 (16.8)	
Onset of seizure				0.027*
Daytime	260 (66.5)	187 (70.3)	73 (58.4)	
Night time	131 (33.5)	79 (29.7)	52 (41.6)	
Etiology of seizure				< 0.001*
Acute symptomatic	183 (44.2)	138 (49.3)	45 (33.6)	
Remote symptomatic	119 (28.7)	63 (22.5)	56 (41.8)	
Cryptogenic	112 (27.1)	79 (28.2)	33 (24.6)	
Investigation(s) at the time of first seizure in ED				
CT scan/MRI				0.074
Normal	144 (38.7)	107 (42.0)	37 (31.6)	
Abnormal	228 (61.3)	148 (58.0)	80 (68.4)	
Lumbar puncture				> 0.999
Normal	33 (55.0)	24 (54.5)	9 (56.2)	
Abnormal	27 (45.0)	20 (45.5)	7 (43.8)	
Antiepileptic drug treatment				< 0.001*
No	137 (33.3)	112 (40.3)	25 (18.7)	
Yes	275 (66.7)	166 (59.7)	109 (81.3)	
Underlying disease				
Stroke[a]				0.001*
No	312 (75.4)	225 (80.4)	87 (64.9)	
Yes	102 (24.6)	55 (19.6)	47 (35.1)	
Essential hypertension				0.659
No	330 (79.7)	221 (78.9)	109 (81.3)	
Yes	84 (20.3)	59 (21.1)	25 (18.7)	
Diabetes mellitus				0.062
No	371 (89.6)	245 (87.5)	126 (94.0)	
Yes	43 (10.4)	35 (12.5)	8 (6.0)	
Cardiovascular disorder[b]				0.308
No	390 (94.2)	261 (93.2)	129 (96.3)	
Yes	24 (5.8)	19 (6.8)	5 (3.7)	
Immune deficiency state[c]				0.759
No	402 (97.1)	271 (96.8)	131 (97.8)	
Yes	12 (2.9)	9 (3.2)	3 (2.2)	
HIV infection				> 0.999
No	402 (97.1)	272 (97.1)	130 (97.0)	
Yes	12 (2.9)	8 (2.9)	4 (3.0)	
Systemic cancer				0.546
No	344 (83.1)	230 (82.1)	114 (85.1)	
Yes	70 (16.9)	50 (17.9)	20 (14.9)	
Medications/substances use				
Alcohol				0.466
No	358 (86.5)	245 (87.5)	113 (84.3)	
Yes	56 (13.5)	35 (12.5)	21 (15.7)	
Antidepressants				> 0.999
No	402 (97.1)	272 (97.1)	130 (97.0)	
Yes	12 (2.9)	8 (2.9)	4 (3.0)	
Antibiotics				0.563
No	400 (96.6)	269 (96.1)	131 (97.8)	
Yes	14 (3.4)	11 (3.9)	3 (2.2)	
Cardiovascular drugs				0.541
No	300 (72.5)	206 (73.6)	94 (70.1)	
Yes	114 (27.5)	74 (26.4)	40 (29.9)	
Systemic chemotherapy				0.607

(*continued on next page*)

Table 1 (*continued*)

Variable	Total N = 414	No seizure recurrence N = 280	Seizure recurrence N = 134	*P* value
No	374 (90.3)	251 (89.6)	123 (91.8)	
Yes	40 (9.7)	29 (10.4)	11 (8.2)	

Data are presented as n (%) unless indicated otherwise.
[a] both ischemic & hemorrhagic stroke.
[b] excluding strokes.
[c] excluding HIV infection.
* Chi-square test, $P < 0.05$.

Table 2
Predictors of first recurrent seizure after the first seizure during overall follow-up time and time of seizure recurrence.

Time of recurrence	Predictors	Crude HR (95% CI)	adj. HR[#] (95% CI)	*P* value
Overall	Nocturnal onset *vs.* daytime	1.42 (0.96, 2.09)	1.53 (1.03, 2.26)	0.039*
	Etiology (ref.: acute symptomatic)			0.003*
	- Remote symptomatic	1.92 (1.29, 3.00)	2.21 (1.38, 3.55)	
	- Cryptogenic	1.21 (0.73, 2.01)	1.71 (0.70, 1.95)	
	Age ≥ 65 *vs.* < 65 y	0.99 (0.98, 1.00)	0.99 (0.98, 1.00)	0.118
0-1 month	Nocturnal onset *vs.* daytime	2.85 (1.48, 5.50)	2.78 (1.44, 5.37)	0.002*
	Antiepileptic drug use (yes *vs.* no)	2.27 (1.03, 4.98)	2.19 (1.00, 4.80)	0.038*
> 1-3 months	Etiology (ref: acute symptomatic)			0.006*
	- Remote symptomatic	3.43 (1.47, 8.01)	3.96 (1.63, 9.60)	
	- Cryptogenic	1.28 (0.45, 3.70)	1.47 (0.50, 4.33)	
	Antiepileptic drug	5.81 (1.76, 19.16)	4.90 (1.42, 16.95)	0.003*
	Age ≥ 65 *vs.* < 65 y	0.98 (0.97,1.00)	0.97 (0.95, 0.99)	0.002*
> 3-12 months	Etiology (ref: acute symptomatic)			0.082
	- Remote symptomatic	2.50 (1.08,5.77)		
	- Cryptogenic	1.91 (0.78,4.70)		

Cox proportional hazard model, $*P < 0.05$.
[#] adjusted for all demographic and clinical characteristics of first seizure.
Abbreviations: HR, hazard ratio; CI, confidence interval; adj. HR, adjusted hazard ratio.

within five years [8–10]. Kim, et al. found that either abnormal neurological signs or symptomatic etiologies in combination with an abnormal EEG showing epileptiform discharges were stronger predictors of seizure recurrence than either of them alone [9]. Another prospective cohort study confirmed the significance of an abnormal CT scan of brain and epileptiform EEG (spikes, polyspikes or sharp activities) in predicting recurrent seizures within two years, whereas advanced age, family history of epilepsy, and seizure types were not associated with recurrence at all [12]. Previous studies which found older age to be a significant predictor of seizure recurrence included only small numbers of elderly patients, possibly resulting in inclusion bias [7,22]. Based on our findings, and a preponderance of evidence from other studies as discussed, we suggest that older age should not be considered as a significant predictor of seizure recurrence. Instead, the presence of neurological deficit, abnormal EEG, and a relevant cerebral abnormality demonstrated by neuro-radiologic studies should be considered as significant predictors for seizure recurrence and epilepsy subsequently.

The advantage of an immediate EEG performed in the ED is obvious. One study demonstrated that up to 21 % of epileptiform discharges were found in the EEGs done within 11 h of FS onset [23], and epileptiform discharges were still detectable in 41 % and 31 % of the EEGs performed within 24 h after the FS and later as a routine re-evaluation, respectively. Our study had a similar finding, as nearly half of the small number of routine EEGs (18 of 35) performed > 24 h after the FS showed epileptiform discharges. A diagnosis of epilepsy cannot totally be excluded by a normal CT scan or EEG, because both studies have low negative predictive values (47 % and 58 %, respectively) for predicting recurrent seizures or epilepsy [12]. To date, the sensitivity and specificity of combined CT scan or MRI of the brain together with an EEG on predicting seizure recurrence have not been extensively investigated. As epilepsy is a neuro-electrophysiological disorder, the usefulness of an EEG in diagnosing a seizure and predicting seizure recurrence (or epilepsy) seems to be more rational than neuroimaging studies. Anatomical lesions on neuroimaging studies are not always epileptogenic foci. It means although there is an apparent lesion, it is not always the etiologic cause of seizure recurrence or epilepsy. Due to the lack of immediate EEG in our ED, the ED physician or neurologist's decision to prescribe an AED to a patient with FS is based mostly on clinical evidence suggesting a high risk of seizure recurrence. Since there were only a very limited number of early EEGs done in our ED during the study period, statistical analysis on this point was impractical.

There was a large difference concerning the effect of AED treatment in reduction of seizure recurrence in the two observational studies noted above, in which a 30 % risk reduction was found in the MESS study, while a 60 % risk reduction was found in the FIR.S.T study. However, when the patients were randomized into "immediate" and "deferred" treatment groups, there was only a modest difference in the reduction of seizure recurrence at 2 years in the treatment arms of both studies. Therefore, the actual benefit in reduction of seizure recurrence risk of immediate treatment with AED in patients with FS remains inconclusive, particularly when long-term follow-ups are considered [24]. In contrast, AED prescription was a significant independent predictor of seizure recurrence in both the early and intermediate recurrence groups in our study. Although AED prescription for patients with FS seems to reduce the risk of recurrence, we believe that the seriousness of the "symptomatic" etiologies (ASS and RSS) correlates with a high risk of seizure recurrence regardless of AED treatment. (Table 1) The prescription of AED in our ED setting is reasonable for short-term prevention of seizure recurrence since the risk of seizure recurrence was perceived as high among our FS patients by a consultant neurologist. Nonetheless, it may not be a significant predictor of long-term seizure recurrence or incipient epilepsy.

The limitation of this study was its retrospective design. However, we aimed to demonstrate the incidence and predictors of recurrent seizures after FS under a specific condition, namely an ED service setting. We hope that the findings of this study will assist the decision-making of ED physicians and consultant neurologists in the management of FS patients in situations with limited EEG availability. Further prospective cohort studies would be very useful in elucidating more fully the nature of seizure recurrence after FS presentation to ED. Future studies to emphasize the application of an immediate EEG in combination with relevant seizure characteristics in predicting seizure recurrence or epilepsy in ED patients who present with FS would be useful in making appropriate management decisions.

5. Conclusion

Though the significant predictors of seizure recurrence were not different from those formerly reported in other settings, the higher incidence and much earlier recurrence among the ED patients with FS in our study warrant a policy of more intensive monitoring of these patients in the future.

Funding and support

There was no funding or any type of non-financial support from any agency

Ethical considerations

The study protocol was reviewed and approved by the Ethics Committee of the Faculty of Medicine, Prince of Songkla University (Registration no. 58-333-14-4). We strictly followed the 1964 Declaration of Helsinki and its later amendments and related good clinical practice guidelines in conducting the research. All information related to patient identity was completely anonymized.

Author contributions

Pornchai Sathirapanya M.D.: Conceptualization of the intellectual content of the study, methodology design, editing the final draft and approving the manuscript before submission

Amika Praipanapong M.D.: Conceptualization of the intellectual content of the study, primary data acquisition and clearance, formal statistical analysis, drafting the original manuscript and approving the manuscript before submission

Chanon Kongkamol M.D.: Performed formal statistical analysis, and approved the manuscript before submission

Pensri Chongphattararot B.Sc.: Performed formal statistical analysis, and approved the manuscript before submission

Declaration of Competing Interest

All authors declare no conflicts of interest related to the research study and manuscript writing.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.seizure.2020.02.007.

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